

GRABOUW ELGIN, South Africa

WMO: 689250

Lat: 34.153S

Lon: 19.030E

Elev: 974

StdP: 14.19

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	36.0	38.2	31.2	26.4	46.8	33.5	29.1	44.6	26.9	57.5	24.4	55.9	0.5	110	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.0	88.6	68.4	84.9	67.3	81.8	66.4	70.6	84.3	69.2	81.3	67.9	78.6	5.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	100.6	75.5	65.0	96.0	74.1	63.8	91.9	72.9	35.1	84.7	33.9	81.4	32.8	79.0	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	16.8	13.6	DB	32.3	97.2	1.1	2.2	31.5	98.8	30.9	100.1	30.2	101.4	29.4	103.0	(4)
(5)			WB	32.0	73.0	1.3	1.0	31.0	73.7	30.2	74.3	29.5	74.9	28.5	75.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	70.0	70.5	67.5	62.2	57.5	53.5	52.9	52.7	55.4	60.2	63.6	67.8	(6)	
	DBStd	8.14	5.24	4.81	5.43	5.52	4.36	4.37	4.83	4.17	4.65	5.52	5.20	5.04	(7)	
	HDD50	67	0	0	0	0	2	15	23	18	8	1	0	0	(8)	
	HDD65	2079	14	8	34	117	238	345	375	380	289	167	85	27	(9)	
	CDD50	4126	621	574	543	366	234	121	113	103	170	319	409	553	(10)	
	CDD65	663	170	163	112	32	4	0	1	0	2	20	45	114	(11)	
	CDH74	3796	971	875	580	181	30	11	17	8	15	164	273	671	(12)	
	CDH80	1081	316	262	172	33	3	0	0	1	2	45	63	184	(13)	
(14) Wind	WSAvg	4.2	3.8	3.6	3.3	3.4	4.2	5.0	4.9	4.9	4.6	4.2	4.1	4.0	(14)	
(15) Precipitation	PrecAvg	50.1	1.5	1.3	1.5	3.6	6.4	8.2	7.4	6.6	4.4	3.9	3.1	2.3	(15)	
	PrecMax	65.2	6.1	3.1	4.2	10.4	11.0	13.9	13.4	14.7	10.4	11.4	8.2	6.7	(16)	
	PrecMin	19.8	0.3	0.2	0.2	0.7	3.2	2.2	0.7	1.7	1.3	0.3	0.7	0.6	(17)	
	PrecStd	9.0	1.3	0.8	0.9	2.1	2.2	2.9	3.0	2.6	2.4	2.5	2.2	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.4	92.1	90.8	84.9	78.1	75.5	76.4	73.8	76.9	87.2	86.7	90.3	(19)	
		MCWB	69.9	69.8	68.9	65.8	60.6	56.6	54.0	55.1	61.2	67.0	65.5	68.4	(20)	
	2%	DB	87.3	86.6	84.5	79.4	72.7	69.1	70.9	67.3	71.0	78.7	80.8	84.8	(21)	
		MCWB	69.0	68.3	67.3	64.7	59.4	54.8	53.7	55.5	59.0	63.4	63.6	67.1	(22)	
	5%	DB	83.1	82.8	80.1	75.3	68.8	64.4	65.5	63.5	66.9	73.8	76.6	80.8	(23)	
		MCWB	67.6	67.5	66.1	63.3	58.5	54.1	53.6	54.5	57.5	61.5	62.8	66.1	(24)	
	10%	DB	79.2	79.3	76.4	71.3	65.6	61.2	61.4	60.3	63.3	69.6	73.0	77.2	(25)	
		MCWB	66.3	66.7	64.7	61.8	57.2	53.4	53.4	53.3	55.5	59.6	61.8	64.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.7	72.4	71.4	68.4	64.2	60.0	59.1	60.2	64.0	68.2	69.1	70.3	(27)	
		MCDB	88.4	85.7	86.8	80.2	72.4	68.4	68.5	67.5	73.7	82.3	81.7	85.6	(28)	
	2%	WB	70.7	70.1	69.0	66.1	61.3	57.4	56.8	57.7	61.0	65.4	66.4	68.3	(29)	
		MCDB	84.6	82.3	81.7	76.1	68.9	63.0	64.6	63.0	68.9	77.1	77.0	81.7	(30)	
	5%	WB	68.9	68.7	67.1	64.4	59.5	55.9	55.1	55.6	58.4	62.7	64.3	66.8	(31)	
		MCDB	80.3	79.5	77.6	73.6	66.7	61.3	62.0	61.0	65.4	71.5	73.3	78.5	(32)	
	10%	WB	67.2	67.4	65.5	62.7	58.1	54.5	53.7	53.8	56.3	60.6	62.4	65.3	(33)	
		MCDB	77.0	76.9	74.3	70.1	64.5	59.7	60.2	58.8	62.3	68.3	70.7	75.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	18.0	18.2	18.0	17.2	16.0	16.4	14.8	15.5	17.0	17.8	18.2	(35)	
		MCDBR	25.1	24.0	24.6	24.3	23.9	23.2	25.1	21.6	22.6	25.3	24.8	24.3	(36)	
	5% WB	MCWBR	9.6	8.8	9.4	10.9	13.3	13.5	13.6	12.4	11.8	11.6	10.9	10.2	(37)	
		MCWBR	22.7	20.9	22.1	22.3	21.3	17.9	20.3	18.4	21.3	23.1	22.2	22.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.333	0.335	0.326	0.312	0.308	0.302	0.303	0.320	0.330	0.341	0.329	0.334	(40)	
		taud	2.550	2.551	2.571	2.606	2.604	2.595	2.582	2.530	2.472	2.458	2.535	2.539	(41)	
	Ebn at Noon		318	311	303	290	274	267	272	282	296	305	318	319	(42)	
		Edh at Noon	34	33	30	26	23	22	23	28	33	36	35	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		2540	2266	1820	1332	946	795	883	1125	1547	2007	2385	2573	(44)	
	RadStd		69	71	80	64	55	46	47	53	68	101	81	83	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (12 neighbors)		+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110	

Nomenclature: See separate page